Project Title:Open Source linux

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Course: Open Source Software

Date of Submission:31 March ,2026

INTRODUCTION

Open Source Software (OSS) brings a significant change in how technology is created. Unlike proprietary software, which keeps its code hidden, OSS focuses on transparency, collaboration, and the freedom to share. At its heart, OSS is about the "Four Freedoms": to run, study, change, and distribute software for any purpose. In today’s computing world, OSS is the unseen foundation of the internet. From the Linux kernels powering servers to the compilers that build our apps, open source prevents any single organization from monopolizing innovation. This report offers a detailed look at VLC Media Player, exploring its origins as a student project at École Centrale Paris, its strict compliance with the GNU General Public License (GPL), and its strong integration within the Linux environment.

ORIGIN AND PHILOSOPHY

Origin Story

In the mid-1990s students at École Centrale Paris had a problem. The campus network was too slow for high-quality video.Before VLC playing videos was messy. People needed a player or "codec" for each file type. Streaming video was also very expensive. Not many people could use it.The idea for VLC started in 1996. It was meant to help stream videos across the university network.The developers made VLC source in 2001. They thought a media player should be free and useful for everyone, not a product to sell.By making it source they let developers all over the world help improve it. Now VLC can play any video format without needing extra downloads.VLC media player became very popular. It is, like a tool that can do things.

License Analysis

The VLC program is mainly licensed under the GNU General Public License. This license is special because if someone changes VLC and shares it with others they have to share their changesThe VLC program has what people call The Four Freedoms:

1. Freedom 0: You can use the VLC program for your use or for business.

2. Freedom 1: You can look at the source code of VLC to see how it works with video files.

3. Freedom 2: You can share the VLC program with your friends.

4. Freedom 3: You can make the VLC program better. Release your own version of VLC.GPL versus MIT license:The MIT license is easy going it lets you take the code change it and sell it without sharing your changes. The GPL license used by the VLC program is different it makes sure the VLC program is always free. Big companies like Apple or Google can use code with the MIT license, in their products without sharing but they cannot do that with the VLC program without sharing their changes with everyone.

The Ethics of Open Source

The ethics of open source software are really about one thing: building on what other people have done. This idea is often called "standing on the shoulders of giants." When developers do not have to start from scratch they can make things faster.For open source software: it stops companies from controlling everything. It lets anyone check the software for security problems. Against open source software: some people think it is not fair to developers because they cannot make money from their work when someone else can just use a free version.The role of companies: companies, like Red Hat and Google make a lot of money by selling help or cloud services that use open source software. This shows that open source software can be free and still make a lot of money for companies that use it. Open source software is a part of what these companies do.

Linux Footprint

VLC is a deal on Linux. It works well with Linux because it uses the usual package managers and it follows the rules of the Filesystem Hierarchy Standard.Installation and management of VLC are pretty easy.On Ubuntu or Debian you can get VLC by using the Advanced Package Tool. You do this by typing: sudo update and then sudo apt install vlcLinux Directory Structure

The binary file is in /usr/bin/vlc. This is the file that actually runs VLC.The libraries are in /usr/lib/vlc. These are the plugins and codecs that VLC needs to work.Your personal settings for VLC are in ~/.config/vlc/. This is where VLC stores the things you like. The global data for VLC is in /usr/share/vlc/. This is where the icons and interface files are.Permissions and updates for VLC are simple.VLC runs with the permissions as a regular user. You do not need to be the boss of the computer to play a video with VLC. This makes it more secure.When it is time to update VLC the system can do it automatically with upgrades.. You can do it manually by typing sudo apt upgrade. VLC is updated along, with the rest of the system.

THE FOSS ECOSYSTEM

The FOSS Ecosystem is really big.VLC is a part of this ecosystem. It does not work alone.The FOSS Ecosystem is like a team and VLC is the heart of this team.Here are a few things about the FOSS Ecosystem and VLC:Dependencies: VLC needs libraries to work properly. For example it needs FFmpeg to decode videos and Qt to make the interface.Tools Built on Top: Many media players on Android and streaming services use the LibVLC engine to process videos.Community: The VideoLAN Organization is a -profit group that manages thousands of volunteer developers. These developers send code fixes to Git. Role in Tech Stack: Server administrators use VLC to change video feeds into webfriendly streams. They use the command line to do this. VLC helps to change the video feed into formats like HLS or DASH. The FOSS Ecosystem and VLC are important, for this.

SCRIPT 1 - SYSTEM IDENTITY REPORT

```
user@ubuntu :~ $ bash sysinfo.sh
--- SYSTEM IDENTIFIED ---
Kernel: 6.5.0-41-generic
User: user
Uptime: up 3 hours, 42 minutes
Date: Tue Mar 31 10:24:37 IST 2026
user@ubuntu :~ $
```

Program Statement: This script displays system details like kernel version, user, and uptime.

Explanation: This script uses variables and system commands like uname, whoami, and uptime to fetch system information.

SCRIPT 2- PACKAGE CHECKER

```
bash — 80×24

ser@ubuntu :~ $ bash pkgcheck.sh
nter package to check: curl
TATUS: curl is installed (Version: 7.88.1-10+deb12u5)

ser@ubuntu :~ $ bash pkgcheck.sh
nter package to check: nmap
TATUS: nmap is NOT found.

ser@ubuntu :~ $
```

Program Statement: Checks whether Apache is installed and displays its details.

Explanation: Uses if-else and case statements along with package management commands.

SCRIPT 3 - DISK AUDITOR

```
bash — 80×24

user@ubuntu :~ $ bash dirinfo.sh

DIR              SIZE             PERM
/etc             12M              755
/home            4.8G             755
/var/log         236M             755

user@ubuntu :~ $
```

Program Statement: Displays directory sizes and permissions.

Explanation: Uses loops and commands like du and ls.

SCRIPT 4 -LOG ANALYZER

```
user@ubuntu :~ $ bash loganalyzer.sh
Log file not found! Create 'syslog.txt' first.

user@ubuntu :~ $ cat syslog.txt
[INFO] Service started successfully
[WARNING] Disk usage above 80%
[ERROR] Failed to connect to database
[WARNING] Memory usage high
[ERROR] Timeout on port 8080

user@ubuntu :~ $ bash loganalyzer.sh
Log Analysis for: syslog.txt
---------------------------
```

Program Statement: Analyzes log files and counts errors.

Explanation: Uses a while loop and condition checking.

SCRIPT 5- MANIFESTO GENERATOR

```
bash — 80×24

user@ubuntu :~ $ bash manifesto.sh
--- FOSS Manifesto Creator ---
Enter your Project Name:   OpenLibre
What is your main mission?  empower users with free and open software
What is your core value?  freedom and transparency
-------------------------------------------
Success! Your manifesto has been saved to 'manifesto.txt'.
Content of file:
PROJECT:   OpenLibre
MISSION:   Our goal is to empower users with free and open software.
PHILOSOPHY:We believe that freedom and transparency is essential for all users.
CREATED BY:user on 2026-03-31

user@ubuntu :~ $
```

```
Log Analysis for: syslog.txt
--------------------------
Total Errors: 2
Total Warnings: 2

user@ubuntu :~ $
```

Program Statement:

Generates a personalized open-source statement.

Explanation:

Uses user input and file handling.

CONCLUSION

This project has helped me comprehend both the technical and philosophical sides of open-source software. I have learned how to use software applications, such as Apache HTTP Server, in the Linux operating system; I have also learned what they provide for modern computing.Developers who use open-source software can collaborate with each other as they develop new technology and learn new things while they do it. Developers can improve upon current technology by creating better solutions using existing technology. This project has allowed me to gain knowledge about scripting, Linux, and the open-source ecosystem.